



Nutrition in the life cycle assessment of foods—function or impact?

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Abstract

Purpose This work provides an unambiguous conceptual framework for inclusion of nutrition in Life Cycle Assessments (LCAs) of food that enables the distinction between two different roles of nutrition, namely serving as the basis of food comparisons via the functional unit and as an impact pathway that links food ingestion with human health effects.

Methods We compare how nutritional aspects have been considered in the functional unit of published LCAs of food with the procedural requirements for ensuring comparability of the functional units. We consider the relevance of nutrient profiling models for assessing food- and diet-related health damages and benefits in the context of LCAs of food. We finally recommend a method that will enable a systematic, comparative, and holistic assessment of the marginal health effect of food products on human health.

Results and discussion Satiety is proposed as a central attribute for comparisons of food products, while weighted measures of nutrient content are suggested to be largely misplaced as part of the functional unit. In contrast, nutritional measures have a large role to play in assessing the human health impacts of the marginal ingestion of specific food products. Such measures should enable a direct quantification of human health effect and benefits and should take advantage of robust epidemiological evidence.

Conclusions Nutritional measures enter into both the functional unit in the form of satiety measures and into the calculation of impacts in the form of the marginal influence of the specific food item on the human health impact of the overall diet. To enhance the differentiation of health impacts at the level of individual food items, it is recommended to combine the nutrient balance indicator with the DALY Nutritional Index (DANI) in each specific dietary context.

Keywords Dietary risk · Functional unit · Health impact · Life cycle assessment · Nutritional quality · Satiety

1 Rationale and objective

The purpose of Life Cycle Assessment (LCA) is to compare the environmental impacts of products, in order to support decisions that can lead to a reduction in the overall impacts. The compared products must fulfil the same needs, i.e. have the same functional unit. Environmental impacts include impacts on resources, ecosystems, and human health. While nutrition is one of the obvious functions of food, a diet that is not nutritionally balanced can cause human health impacts. This

dual role of nutrition has led to some confusion on how to include nutritional measures in LCA. The purpose of this work is to provide an unambiguous framework for inclusion of nutrition in LCAs of food, clearly distinguishing the role of nutrition in the functional unit and in the impact calculations.

2 Approach and methodology

We compare how nutritional aspects have been considered in the functional unit of LCAs of food with the procedural requirements for ensuring comparability of the functional units. We likewise consider the relevance of nutrient profiling models for assessing food-related health impacts. We identify the nutritional measures of food intake that should be included in LCAs of food, in order to provide a systematic, comparative, and holistic assessment of the marginal impact of specific food products on human health, i.e. the impact of the specific food being added to or subtracted from an existing diet.

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3 Results and discussion

3.1 Nutrition as a function

Nutrition can be seen as the “original” biological function of food. With this starting point, many researchers have suggested that nutrient content should be included when determining the functional unit of food (Heller et al. 2013). The most simple and common approaches rely on a single component of nutrition, such as metabolic energy or protein (e.g. Sonesson et al. 2017). More complex functional units have been suggested that rely on nutritional profiling algorithms that summarize in a single score weighted measures of qualifying and/or disqualifying nutrients based on pre-existing nutrition guidelines, recommendations, or healthy dietary patterns, e.g. the Weighted Nutrient Density Score (WNDS) of Arsenault et al. (2012). Drewnowski and Fulgoni (2008) reviewed a large number of such nutrient profiling models and van Dooren (2016) noted that they all have the same structure and that it was sufficient to include three nutrients, namely total protein, essential fatty acids, and dietary fibre, because these elements correlate significantly with all other essential nutrients. In van Dooren’s resulting Nutrient Density Unit, the limiting macronutrients were left out because they can result in a functional unit with negative values. A negative functional unit obviously creates a conceptual problem for LCA, as also pointed out by Heller et al. (2013). This is a sign of a more fundamental problem of seeking to include in the functional unit the potential health damages related to nutrition. Bruun Werner et al. (2014) and van Dooren et al. (2017) even go a step further and include the greenhouse gas emissions into an overall “Sustainable Nutrient Rich Food index”, thus reflecting a weighted index of global warming and nutritional impacts relative to the energy density of the food products. Since LCA results are normally expressed as impacts per functional unit, the inclusion of the most important impacts in the denominator leaves the question: *What is then left in the numerator?*

For a product property—like nutrition in foods—to be relevant from the perspective of the functional unit, it must be obligatory; i.e. it must be essential for the product to be considered as a relevant alternative by the customers (Weidema 2017). Specific *desirable* nutrients only occur as obligatory properties in purchase decisions for “functional foods” and in situations where food is insufficiently accessible. In all other purchase situations, the desirable nutrients are not obligatory properties; i.e. they are not essential for a specific food to be considered as an acceptable alternative by customers. Desirable nutrients are rather positioning product properties, similar to price, that may place a food product more or less favourable in comparison. In contrast, specific *undesirable* components, such as allergens, sodium, and certain types of fat, may appear as obligatory properties in the form of

requirements of low content or complete absence of these components.

Beyond nutritional composition, different food components deliver different degrees of satiety (feeling of fullness). When comparing foods with the same energy content, foods rich in protein and fibre deliver more satiety than carbohydrate-rich foods, while sweet-tasting foods and foods rich in fat deliver less satiety (Chambers et al. 2015; Hopkins et al. 2016). Satiety is therefore a central nutritional measure for the functional unit of food products, because it determines how much time will pass before an additional meal or snack will be demanded and/or how much will subsequently be consumed. Even in situations where food is insufficiently accessible, satiety is a relevant measure to include, considering that the purpose of LCA is to compare the products that are actually being substituted, rather than an idealized, abstract measure that is not possible to take directly into account in the purchase situation. Unfortunately, the knowledge on satiety differences between major food groups is not sufficient to establish general quantified measures at the level of individual food components, where results are complicated by synergy effects and dependencies on temperature, palatability, and more detailed chemical properties, such as stereochemistry in sugars. Therefore currently, for individual foods, differences in satiety can only be established for foods that have been compared specifically under appropriate experimental conditions. In the absence of such relevant experiments, the general knowledge on satiety differences therefore mainly serves as a warning of too simplified comparisons of foods based on their energy content alone.

So to conclude on how nutrition should enter into the functional unit, we can express this in the often used “how much, how long, how well” terms from the ILCD Handbook (IES 2010):

- How much? Portion size
- How long? Satiety-weight
- How well? Exclusion criteria or min/max intervals

but generally *not* as:

- Continuous variables of specific nutrients
- Simple mass of product (often used with the excuse that the uncertainty on the above-mentioned functional unit components are too high, which should rather be an argument for providing uncertainty information on these variables).

3.2 Nutrition as a risk factor for human health

A nutritionally balanced diet is a prerequisite for human health. There is a rather large span within which a diet can

be regarded as balanced, and within which it is considered that there is no increased risk of developing an adverse health effect by an addition or subtraction of a specific dietary risk (food group, food item, or nutrient). However, for a diet that is already unbalanced, i.e. already above or below “minimum risk” intake limits for detrimental and beneficial dietary risks, respectively, the addition or subtraction of a specific food item will have a marginal influence on related dietary risks and thus can lead to beneficial or detrimental human health effects. Dietary risks from an unbalanced diet constitute the single most important cluster of risk factor for preventable human health impacts causing 9.6% of the 2016 Global Burden of Disease or 229 million disability-adjusted life-years annually (Forouzanfar et al. 2016).

More than half of the global population lives on an unbalanced diet (Imamura et al. 2015). An unbalanced diet is here understood as a diet low in fruits, vegetables, legumes, whole grains, nuts and seeds, milk, fiber, calcium, polyunsaturated fatty acids, and omega-3 fatty acids or precursors or a diet high in red meat, processed meat, sugar-sweetened beverages, trans fatty acids, and sodium (Gakidou et al. 2017). Specific thresholds and characterization factors for each of these components are provided by the Global Burden of Disease (GBD) studies in the form of the theoretical minimum risk exposure levels (the intake range which poses no added risk) and age-specific risk ratios based on a meta-analysis of epidemiological studies (Forouzanfar et al. 2016; Gakidou et al. 2017). For nutrients, the minimum risk thresholds recommended by the GBD studies are in relative agreement with recommended or maximum daily intake used in the nutrient profiling models described above. For the remaining gross food groups in the dietary assessment of the GBD studies, nutrient profiling models typically include a wider range of nutrients (e.g. vitamins, trace elements, and fatty acids), which can provide a more differentiated quality score to the individual components of the gross food groups. However, nutrient profiling models are only indirectly associated with health effect (McCullough and Willett 2006), limiting their ability to quantify health damages in LCAs of foods and diets. Thus, any combination of nutrient profiling models with the epidemiological data of gross food groups from the GBD studies must be done with great care.

Intake of food is part of the use stage of a food LCA and takes place in the form of either snacking, home-cooked meals, restaurant meals, or institutional meals. For the assessment of the marginal (small, additional) contribution to human health impacts from a marginal change in the intake of an individual food, it is obviously necessary to take into account the large variation in diets between individuals and between specific population groups (Imamura et al. 2015). Most nutrient profiling models are missing considerations of how nutrients interact within the individual foods or the remaining diet (Arvaniti and Panagiotakos 2008). The approach of Fern et al. (2015) seeks to overcome this limitation by adding a nutrient

balance indicator showing the extent to which a food, meal, or diet can satisfy the daily requirements for all qualifying nutrients. In line with van Kernebeek et al. (2013), the nutrient balance indicator of Fern et al. (2015) excludes contributions of qualifying nutrients once the requirement for a specific qualifying nutrient is met, thus preventing inflated values for a food containing abnormally high amounts of one or two individual qualifying nutrients.

An ideal approach to associate foods, meals, or diets with health damages within an LCA context is to utilize epidemiological evidence, like in the GBD studies. A first attempt to implement this approach was presented by Stylianou et al. (2016) where the authors proposed the Combined Nutritional and Environmental Life Cycle Assessment (CONE-LCA) framework that introduces a nutritional impact pathway linking foods and diets to human health damages and benefits. The authors demonstrated how this approach could be implemented in single-component foods, like milk, using a single dietary risk factor to develop a nutritional characterization factor. To be able to utilize this approach to any food or diet, more dietary risk factors need to be considered and combined. However, due to critical differences in epidemiological studies that could influence the quality of data, e.g. study design, risk definition, and assessment method, study population, and consideration of confounding variables, it is important to use pooled estimates derived from consensus data. We consider the work by the GBD studies as that ideal source of such data, since they are among the best recognized international efforts for systematically quantifying multiple risk factors in a comparative and comprehensive manner. However, it should be emphasized that the GBD studies focus on population-level estimates of health burden associated with dietary risks at national, regional, and global levels. Therefore, the GBD estimates cannot be easily used for nutritional health inferences on individual food items in LCA studies.

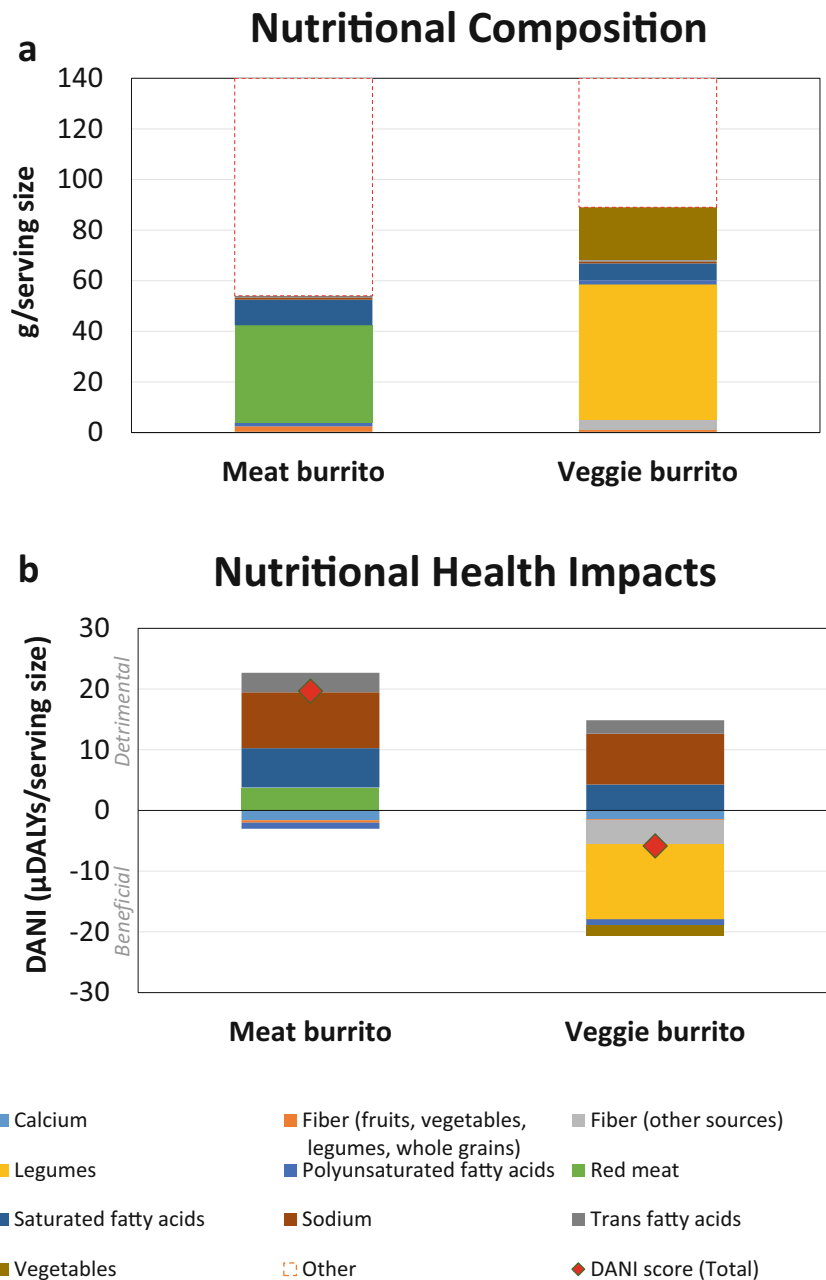
Building on the GBD studies, the DALY Nutritional Index (DANI) offers a new nutritional health assessment method that is conceptually aligned with LCA (Stylianou 2018; Stylianou et al. 2019). More specifically, DANI provides a continuous single-score quantification, expressed in disability-adjusted life years (DALYs) per functional unit of food, of the associated marginal (small, additional) health burden from all-cause premature mortality and disease morbidity, based on 15 dietary risks from the GBD studies plus saturated fatty acids. There are two sets of data required to estimate DANI. First, inventory flows for the 16 dietary risk components need to be determined for the food under study. Fulgoni et al. (2018) have developed a methodology to obtain such nutritional inventory flows for foods in the “What We Eat in America/National Health and Nutrition Examination Surveys” (WWEIA/NHANES) database. These inventory flows quantify the mass of the 16 dietary risk components

present in a food (in $\text{kg}_{\text{risk component}}$) per serving size, in accordance with the GBD definition of the risk, using publicly available nutritional datasets (see example in Fig. 1a). The second set of data needed to estimate DANI scores are nutritional characterization factors for the 16 dietary risk components that measure the marginal change (detrimental or beneficial) in the burden of disease associated with an additional intake of a specific risk component, expressed in $\text{DALYs/kg-risk component}$. Such characterization factors have been developed for US adults (25 years and older), based on epidemiological evidence from 6195 risk-outcome-strata-age group-gender-burden pairs and US-specific burden rates obtained from the GBD studies (Stylianou 2018; Stylianou et al.

2019). A positive characterization factor indicates a risk of detrimental health effects whereas a negative characterization factor indicates a protective health effect (i.e. a reduced health risk). Within a marginal context, i.e. looking at small changes in food intake, which is suitable when evaluating and comparing food items and substitutions, dietary risks are assumed to have an independent and additive effect on health. Hence, marginal DANI scores can then be estimated multiplying these inventory flows by the corresponding nutritional characterization factors.

We demonstrate DANI using an example of two distinct types of burritos in the US diet. Figure 1a presents the nutritional inventory flows for a beef and a bean burrito per standard

Fig. 1 Nutritional inventory flows (a) and corresponding health impacts (b) of a meat and veggie burrito



serving (140 g), obtained from Fulgoni et al. (2018). The composition of this beef and bean burrito covers seven and nine out of the 16 dietary risks in DANI, respectively. As indicated by the inventory flows, the meat burrito was primarily comprised of red meat (7%) and ingredients that are assumed to have neutral effect on health (61%), e.g. poultry and white rice. The veggie burrito predominantly consisted of legumes (38%), vegetables (15%), and ingredients with neutral health effect (36%), e.g. white rice. To avoid double counting the health benefit of fiber on cardiovascular diseases, we distinguish between fibers obtained from “fruit, vegetable, legumes, and whole grains” and “other sources” with the former only benefitting colorectal cancer and the latter benefitting colorectal cancer and ischemic heart disease (Gakidou et al. 2017). The corresponding DANI scores for these food items are estimated at 19.6 and $-5.8 \mu\text{DALYs/serving}$ (Fig. 1B). These scores indicate that an addition of these food items to the average US diet would induce health damages in the case of the meat burrito whereas the veggie burrito has an overall proactive health effect. Health damages associated with the meat burrito are driven by the detrimental health effects of sodium ($9.2 \mu\text{DALYs/serving}$), saturated fatty acids ($6.5 \mu\text{DALYs/serving}$), and to a lesser extent red meat ($3.8 \mu\text{DALYs/serving}$). The veggie burrito was associated with slightly lower detrimental health impacts from sodium and saturated fatty acids compared with the meat burrito, and the health benefits, mainly from legumes ($-12.4 \mu\text{DALYs/serving}$), exceeded any detrimental impacts. This example illustrates that DANI can be used as a nutritional health damage assessment method that is compatible with LCA.

Although this approach is an important step towards facilitating nutrition as an impact pathway for food LCA, there are still several limitations to be addressed. In order to link foods, nutrients, and diets to health damages, we need to rely on the availability, robustness, quality, and granularity of epidemiological evidence. Since it is typical to investigate food groups rather than individual food items to obtain sufficient statistical power, we lack epidemiological data supporting differentiated health effects between foods of the same food group. For example, while there is solid data associating consumption and health effects for sugar-sweetened beverages (SSBs) as a food group, the association only supports a dose-response relationship based on the amount of beverage consumed, while associations based on the sugar content of the beverages either do not exist or are not robust. Thus, the epidemiological data does not allow extrapolation to the health effects of added sugar in general. As a result, when two SSBs of different sugar content are compared, the use of epidemiological evidence would support the same health effect for both products while the use of nutritional profiling models that rely on the nutrient content of the food would result in different scores. In this specific example, it appears warranted to accept extrapolations to added sugar in general, i.e. being less demanding on the epidemiological evidence for individual food items.

Another problematic issue is the generalization of the epidemiological evidence from the study to the general population. The robust associations between dietary risks and health outcome that typically originate from studies in Europe and the USA may not be valid for other populations. Such evidence starts emerging from studies in Asia that have found a beneficial health effect associated with increase red meat consumption (e.g. Lee et al. 2013) whereas the consensus from European and US studies (that are often applied to other regions) supports an inverse association (Forouzanfar et al. 2016).

Advancements in nutrition research support that nutritional assessment methods should focus more on food groups and dietary patterns—that provide evidence on the aggregated effect of nutrients—and less on nutrients (Mozaffarian 2017). While DANI follows this approach, in its current form (that evaluates the health effect of a marginal dietary change at the food level), it fails to capture the aggregated health effect of nutrients and food groups at the diet level, which are multiplicative and not additive as in the marginal dietary change scenario. Thus, using marginal DANI might lead to overestimating the nutritional benefits of “healthy” diets and underestimate the benefits from “unhealthy” diets, and extrapolating health risk to individual food items might be pushing the boundaries of epidemiological information. However, food-level health burden comparisons are practical and useful, as they can help with the identification of foods to increase or to reduce in the diet, providing a quick way to motivate and make informed decisions, similar to food labelling (Sacks et al. 2009). Such an approach can address challenges of dietary recommendations in achieving healthy and sustainable transitions, since diet-level information is often complex and hard to convey in a clear actionable manner (Beales et al. 2004). Finally, more recent efforts focus on marginal food-level improvements, since they can be more realistic, be maintained in the long term and generate substantial health benefits.

Although nutritional impact assessment in food LCA is at its infancy, adaptations of this approach by LCA practitioners can lead to improved and more comprehensive evaluation of foods and diet. For food comparisons, marginal DANI should be treated as a relative comparison measure, which is in line with the LCA framework, and all associated limitations should be acknowledged. Ideally, the health impact associated with specific food items should be determined as the health burden change that results in from the diets with and without the food, with multiple food comparisons assuming the same diets. This adds complexity to the evaluation of the health impact characterization for LCA, since the definition and evaluation of the diet are required. Yet, this is possible and forthcoming. Future improvements of nutritional impact assessment in LCA include the development of a diet-DANI that accounts for multiplicative aggregated health effects of dietary risks and is complemented with the nutrient balance indicator of Fern et al. (2015).

So to conclude on how nutrition should enter into the impact assessment, we can use nutritional assessment tools that link food, nutrients, and diets with human health effects that take advantage of robust epidemiological evidence:

- The GBD studies offer the best available consensus on epidemiological evidence for dietary risk factors.
- Nutritional inventory flows need to be aligned with the definitions of the dietary risk factors as in e.g. Fulgoni et al. (2018).
- The contribution of components should be weighted based on their effect magnitude, and prevalence and severity of the health outcomes (e.g. the nutritional characterization factors used in DANI)

4 Conclusions

To answer the question raised in the title, it is not an either/or, but rather that nutritional measures enter into both the functional unit in the form of satiety measures and into the calculation of impacts in the form of the marginal (small, additional) influence of the specific food on the human health impact of the overall diet.

The GBD health risk ratios for each qualifying and disqualifying food group or nutrient can provide measures of health impact per marginal amount of intake in each specific dietary context. Consequently, the DANI is a novel and promising approach that translates the dietary risks in the GBD to health burden scores in DALYs/FU, using the nutritional composition of food items. The use of DANI in food LCAs introduces a nutritional impact category in the life cycle impact assessment framework and enables the quantification of the trade-offs between environmental and nutritional health damages and a more comprehensive assessment of foods. DANI may be combined with the nutrient balance indicator to enhance the differentiation at the level of individual foods.

Both satiety and marginal health impacts of food intake are complex mechanisms with a large variation between individuals and population groups. However, this should not be an argument for their exclusion from LCAs of food products, but rather for the inclusion of realistic uncertainty measures for each of the mechanisms involved.

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